Highlights

Log-Weighted Cross-Entropy Framework for Imbalanced Classification

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- A logarithmic re-weighting loss that bounds the weight explosion of weighted cross-entropy.
- A single-parameter generalization tuned automatically on a small proxy subset.
- An ε-smoothed variant keeps per-class weights bounded and well defined for the rarest classes.
- Evaluation spans image, tabular, and network-intrusion domains of varying imbalance.
- The proposed losses improve as imbalance grows and avoid the severe degradation of competing re-weighting losses.

Log-Weighted Cross-Entropy Framework for Imbalanced Classification

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ABSTRACT

Class imbalance is a pervasive obstacle in machine learning, arising whenever a few majority categories dominate the training data while critically important minority categories remain severely underrepresented. Among algorithm-level remedies, Weighted Cross-Entropy (WCE) is the most direct: it scales the loss of each class by the inverse of its frequency. However, these inverse-frequency weights grow without bound as the imbalance becomes more severe, producing unstable optimization. More elaborate alternatives such as Class-Balanced (CB) Loss and Focal Loss introduce additional hyperparameters and, in our experiments, behave inconsistently across datasets—occasionally falling below ordinary Cross-Entropy (CE). We propose Log-Weighted Cross-Entropy (LWCE), which replaces the inverse-frequency weight with a logarithmically compressed counterpart that keeps a meaningful boost for rare classes while bounding weight growth, and which introduces no additional hyperparameters. We further generalize it to Power-Log Weighted Cross-Entropy (PLWCE) through a single exponent that adjusts the focusing strength and is selected automatically on a small proxy subset, and we introduce a complementary ϵ-smoothed variant (ES-LWCE) that adds a small constant to the logarithmic denominator, providing an explicit, user-controlled cap on the per-class weight and keeping it well defined even for vanishing class counts, such as classes absent from a mini-batch. To assess generality, we evaluate the proposed losses across three distinct application domains that together span a wide range of imbalance severity: long-tailed image classification, tabular classification benchmarks, and network intrusion detection. Across these domains, LWCE and PLWCE improve over Cross-Entropy as the imbalance grows and, unlike the competing re-weighting losses, avoid the severe degradation below plain Cross-Entropy that these baselines occasionally exhibit. These results indicate that logarithmic weight compression offers a simple, stable, and parameter-efficient alternative to existing re-weighting strategies for class-imbalanced classification.

1. Introduction

Across a wide range of real-world classification tasks—including medical diagnosis, financial fraud detection, network intrusion detection, and visual recognition—the categories of interest are rarely observed in equal proportions. Instead, a small number of majority categories concentrate the overwhelming majority of the available samples, while many minority categories are represented by only a handful of examples (He and Garcia, 2009; Johnson and Khoshgoftaar, 2019). This long-tailed structure is not an artifact of poor data collection but an intrinsic property of the underlying phenomena: healthy patients vastly outnumber those with a rare disease, legitimate transactions dwarf fraudulent ones, and benign network traffic far exceeds malicious traffic. Because the minority categories are often precisely the ones that carry the greatest practical cost when missed, building classifiers that remain reliable under imbalance is a problem of broad and lasting importance.

The central difficulty is that the standard training objective is itself biased toward the majority. When a model is trained with ordinary Cross-Entropy (CE) loss, the aggregate gradient signal is dominated by the numerically abundant classes, so the optimizer can drive the overall loss down while largely ignoring rare categories. The resulting classifier attains a high overall accuracy yet performs poorly on exactly the minority classes that matter most—a failure mode commonly described as majority bias. Standard accuracy therefore masks the problem rather than revealing it, and a method that addresses imbalance must improve minority-class performance without sacrificing the majority.

Existing remedies fall broadly into two families. Resampling methods reshape the training distribution directly: oversampling techniques such as the Synthetic Minority Over-sampling Technique (SMOTE) synthesize new minority

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instances (Chawla, Bowyer, Hall and Kegelmeyer, 2002), while undersampling discards majority examples to restore balance. Both directions carry well-known risks—oversampling can introduce synthetic artifacts and encourage overfitting, whereas undersampling discards potentially informative majority data (Drummond and Holte, 2003). Algorithm-level methods instead modify the training objective. These include margin-based losses that enlarge the decision boundary around minority classes (Cao, Wei, Gaidon, Aréchiga and Ma, 2019), logit-adjustment techniques that calibrate class priors, and loss re-weighting methods that scale the per-class loss by class-dependent weights. Among these, re-weighting is especially attractive for three practical reasons. Unlike resampling, it leaves the training data untouched and therefore introduces neither synthetic artifacts nor information loss; unlike architecture-level remedies, it adds no parameters and can be dropped into virtually any network without altering the model; and it requires only a minimal, modality-independent change to the loss function. Indeed, although many sophisticated long-tailed learning methods have since been proposed, re-weighting remains one of the few approaches that can be applied to almost any architecture without modifying the model itself. For these reasons this paper focuses on the re-weighting family.

Despite its appeal, re-weighting remains unsatisfying in practice. Weighted Cross-Entropy (WCE), the most direct re-weighting scheme, assigns each class a weight inversely proportional to its frequency,

$$w_c^{\mathrm{WCE}} \propto \frac{1}{n_c}. \tag{1}$$

As a minority class becomes rarer and n_c shrinks, this inverse-frequency weight grows without bound relative to the majority, so the corresponding gradients become disproportionately large and destabilize optimization under extreme imbalance. More sophisticated alternatives were designed to soften this behavior but bring their own drawbacks. Focal Loss reduces the influence of easy, typically majority, examples through a tunable focusing term, but it operates on per-sample confidence rather than class frequency and therefore behaves inconsistently across imbalance levels. Class-Balanced (CB) Loss re-weights by an effective number of samples governed by an additional hyperparameter, yet that hyperparameter is sensitive and, in our experiments, CB Loss degrades below plain Cross-Entropy under severe imbalance. Existing methods thus attempt to alleviate this instability through additional mechanisms, yet none directly addresses the uncontrolled growth of the inverse-frequency weight while preserving the simplicity of WCE. In short, the field still lacks a re-weighting loss that is simultaneously simple, hyperparameter-light, numerically stable, and dependable across both the level of imbalance and the application domain.

To address this gap, we propose Log-Weighted Cross-Entropy (LWCE). The key idea is to replace the inverse-frequency weight of WCE with a logarithmically compressed counterpart, applying a logarithm to the class count before inverting it. This preserves a meaningful emphasis on rare classes—the weight still decreases with class size—but fundamentally bounds how fast that emphasis can grow, so the explosive behavior of WCE is removed. Crucially, the resulting loss introduces no additional hyperparameters and can be used as a drop-in replacement for Cross-Entropy. We further develop LWCE in two complementary directions. The first, Power-Log Weighted Cross-Entropy (PLWCE), controls *how strongly* rare classes are emphasized: it introduces a single exponent that continuously adjusts the focusing strength, ranging from near-uniform weighting to an aggressive emphasis on the tail. This exponent is the only free parameter, and it is selected automatically through a lightweight grid search on a small proxy subset, adding negligible computational overhead. The second, ε-Smoothed Log-Weighted Cross-Entropy (ES-LWCE), instead adds a small constant ε to the logarithmic denominator. The base LWCE weight is already bounded for every class that appears at least once, so the role of ES-LWCE is not to make an otherwise unbounded weight finite, but to provide an *explicit, user-controlled* cap of $1/\varepsilon$ on the per-class weight and to keep it well defined in the degenerate case where a class is temporarily absent—for example, when class frequencies are estimated per mini-batch—without changing the relative ordering of class weights. The two extensions are therefore orthogonal: PLWCE adjusts the *strength* of the re-weighting, whereas ES-LWCE controls its *maximum weight* and smooths the extreme tail. Although the two can in principle be combined into a single power-log-smoothed weight, we study them separately to isolate the effect of focusing strength from that of smoothing.

We assess the proposed losses through a deliberately broad empirical study rather than a single benchmark. To test whether the benefit of logarithmic compression generalizes, we evaluate across three application domains that together cover a wide spectrum of imbalance severity: long-tailed image classification on standard vision benchmarks, a collection of tabular classification datasets drawn from public repositories, and several network intrusion detection datasets in which rare attack types are extremely scarce. In every domain we compare the same set of re-weighting losses under a controlled protocol with repeated runs, and we report metrics that explicitly reflect minority-class performance rather than overall accuracy alone. This cross-domain design lets us examine not only whether the proposed losses help, but how their advantage changes as imbalance becomes more extreme.

The main contributions of this paper are as follows.

1. We propose Log-Weighted Cross-Entropy (LWCE), a parameter-free drop-in replacement for Weighted Cross-Entropy that substitutes the inverse-frequency weight with a logarithmically compressed one, removing the weight-explosion failure mode while retaining a meaningful emphasis on rare classes.
2. We develop two complementary extensions of LWCE: Power-Log Weighted Cross-Entropy (PLWCE), whose single exponent—tuned automatically on a small proxy subset—controls the focusing strength, and ϵ-Smoothed Log-Weighted Cross-Entropy (ES-LWCE), whose smoothing constant imposes an explicit, user-controlled cap on the per-class weight and keeps it well defined even for vanishing or mini-batch-level class counts.
3. We analyze the theoretical properties of the proposed weighting functions, showing that their rare-to-frequent weight ratio grows only logarithmically with the imbalance ratio—in contrast to the linear growth of WCE—and that the resulting weights remain bounded and numerically stable even for the rarest classes.
4. We provide a controlled cross-domain evaluation spanning image, tabular, and network intrusion detection data, showing that the proposed losses improve over Cross-Entropy as imbalance grows and, unlike Class-Balanced Loss and Focal Loss, avoid the severe degradation below plain Cross-Entropy that these baselines occasionally exhibit.

The remainder of this paper is organized as follows. Section 2 reviews related work on learning from imbalanced and long-tailed data. Section 3 defines the proposed losses and analyzes their properties. Section 4 describes the experimental setup across the three domains and presents the results. Section 5 concludes and discusses limitations and future work.

2. Related Work

Methods for learning under class imbalance can be organized by the level at which the imbalance is addressed: the *data*, by reshaping the training distribution; the *loss*, by making the training objective cost-sensitive; the *representation*, by rebalancing the learned feature space or classifier; and the *architecture*, by restructuring the model itself (He and Garcia, 2009; Johnson and Khoshgoftaar, 2019; Krawczyk, 2016; Branco, Torgo and Ribeiro, 2016; Zhang, Kang, Hooi, Yan and Feng, 2023; Chen, Yang, Yu, Shi and Chen, 2024). Table 1 summarizes this taxonomy with representative methods. Our proposal is a loss-level method and, more specifically, a frequency-based re-weighting loss. We review all four levels for context and then position logarithmic re-weighting among existing cost-sensitive losses.

Table 1
A taxonomy of approaches to class-imbalanced learning, organized by the level at which the imbalance is addressed. The proposed losses are loss-level, frequency-based re-weighting methods.

Level	Representative methods
Data-level	SMOTE, ADASYN, under-/over-sampling
Loss-level	WCE, CB Loss, Focal Loss, **LWCE (ours)**
Representation-level	Decoupled training, distribution alignment
Architecture-level	Multi-branch, multi-expert models

2.1. Data-level resampling

Resampling rebalances the class distribution before or during training. Oversampling synthesizes additional minority instances—the Synthetic Minority Over-sampling Technique (SMOTE) interpolates between neighbouring minority samples (Chawla et al., 2002), and ADASYN adapts the amount of synthesis to local difficulty (He, Bai, Garcia and Li, 2008)—whereas undersampling discards majority examples and can outperform oversampling on classical learners (Drummond and Holte, 2003). Although effective and classifier-agnostic, synthetic oversampling may introduce noisy or unrealistic samples and encourage overfitting, while undersampling discards potentially informative data (Buda, Maki and Mazurowski, 2018; Branco et al., 2016). These trade-offs motivate objective-level alternatives that leave the data untouched.

2.2. Cost-sensitive and re-weighting losses

Algorithm-level methods instead make the loss itself imbalance-aware. We group them by the signal that drives the re-weighting into frequency-based, difficulty-based, and margin- or logit-based methods. *Frequency-based* methods scale each class loss by a function of its count. The most direct instance is Weighted Cross-Entropy, which sets $w_c \propto 1/n_c$ so that the weight is inversely proportional to class count (Wang, Liu, Wu, Cao, Meng and Chen, 2019); these weights, however, grow without bound under extreme imbalance and destabilize optimization, and the benefit of such importance weighting is known to diminish for over-parameterized networks driven to low training error (Byrd and Lipton, 2019). Class-Balanced Loss softens this by weighting with an effective number of samples, $w_c \propto (1-\beta)/(1-\beta^{n_c})$, but this introduces an extra hyperparameter $\beta \in [0, 1)$ (Cui, Jia, Lin, Song and Belongie, 2019). *Difficulty-based* methods modulate each example by its own hardness rather than by its class: Focal Loss down-weights easy, mostly majority, samples by a factor $(1-p_t)^\gamma$, where p_t is the predicted probability of the true class and γ a tunable focusing exponent (Lin, Goyal, Girshick, He and Dollár, 2017), and the Gradient Harmonizing Mechanism flattens the distribution of gradient norms so that both easy samples and hard outliers are suppressed (Li, Liu and Wang, 2019). *Margin- and logit-based* methods reshape the decision geometry rather than the per-class loss magnitude: they enlarge minority margins (Cao et al., 2019), equalize gradients across frequencies (Tan, Wang, Li, Li, Ouyang, Yan and Yan, 2020), adjust logits by class priors (Menon, Jayasumana, Rawat, Jain, Veit and Kumar, 2021), or learn the weights themselves through meta-learning and cost-sensitive representation learning (Ren, Zeng, Yang and Urtasun, 2018; Ren, Yu, Sheng, Ma, Zhao, Yi and Li, 2020; Khan, Hayat, Bennamoun, Sohel and Togneri, 2018), with recent work casting weight learning as an optimal-transport problem between the imbalanced and a balanced distribution (Guo, Li, Zheng, Zhao, Zhou and Zha, 2022). These losses improve over plain Cross-Entropy in many settings, but they introduce additional, often sensitive, hyperparameters and can behave inconsistently across datasets. Our logarithmic family belongs to the frequency-based group and is therefore complementary to the difficulty- and margin-based axes; it targets the central weakness of frequency-based re-weighting by bounding the frequency weight without introducing sensitive hyperparameters.

2.3. Representation learning and decoupling

A complementary line of work argues that imbalance harms the classifier more than the feature extractor. Decoupling representation and classifier learning (Kang, Xie, Rohrbach, Yan, Gordo, Feng and Kalantidis, 2020) and distribution alignment (Zhang, Li, Yan, He and Sun, 2021) rebalance only at the classification stage, while open long-tailed recognition addresses imbalance jointly with open-set and few-shot effects (Liu, Miao, Zhan, Wang, Gong and Yu, 2019); recent surveys provide a broad taxonomy of these directions (Zhang et al., 2023). This concern extends to the era of large pretrained and foundation models, where the fine-tuning strategy itself interacts with the long tail and must be adapted to avoid harming the rarest classes (Shi, Wei, Zhou, Shao, Han and Li, 2024). Such methods are largely orthogonal to the choice of loss and can be combined with re-weighting, so our contribution—confined to the loss term—composes with them rather than competing.

2.4. Architecture-level and multi-expert methods

A third family addresses imbalance by restructuring the network rather than the data or the loss. Multi-branch designs train separate pathways for the head and tail of the distribution and fuse them through cumulative, bilateral learning (Zhou, Cui, Wei and Chen, 2020). Mixture-of-experts and multi-expert approaches extend this idea by training several diverse, distribution-aware experts and then routing or aggregating their predictions: RIDE routes inputs to complementary experts (Wang, Lian, Miao, Liu and Yu, 2021), ACE assembles ally-complementary experts in a single stage (Cai, Wang and Hwang, 2021), LFME distils knowledge from multiple experts trained on balanced subsets (Xiang, Ding and Han, 2020), and test-agnostic aggregation adapts the expert ensemble to an unknown test distribution (Zhang, Hooi, Hong and Feng, 2022). Such designs can deliver strong tail accuracy, but they increase parameter count and inference cost, and—like the resampling and representation methods above—they are orthogonal to the training objective: a re-weighting loss such as ours can be applied within each branch or expert rather than replacing the architecture.

2.5. Positioning of the proposed losses

Within the re-weighting family, existing schemes face a dilemma: inverse-frequency weighting is parameter-free but explodes, whereas more stable variants regain control only by adding sensitive hyperparameters (Lin et al., 2017; Cui et al., 2019). Logarithmic compression offers a middle ground that, to our knowledge, has not been systematically studied as a re-weighting principle across application domains. LWCE realizes it without any additional hyperparameter, PLWCE adds a single exponent that tunes the focusing strength, and ES-LWCE adds an ε term that keeps the weight bounded and well defined for the rarest classes. Table 2 contrasts these properties with the main cost-sensitive baselines: WCE is parameter-free but its weights are unbounded, Focal and CB Loss regain stability only at the cost of an extra hyperparameter, whereas LWCE is a frequency-based, parameter-free scheme whose weights stay bounded. We evaluate this family against the representative cost-sensitive losses above under a unified protocol spanning image, tabular, and network-intrusion domains—each of which sustains its own active line of imbalance research, from dedicated tabular benchmarks (Liu, Li, Yang, Wei, Kang, Zhu, Hamann, He and Tong, 2025) to intrusion detection under severe class skew (Abdelkhalek and Mashaly, 2023).

Table 2
Positioning of LWCE against representative cost-sensitive losses. "Frequency"/"Difficulty" denote the re-weighting signal; "Param.-free" means no loss hyperparameter is required; "Bounded weight" refers to whether the multiplicative weighting factor is prevented from unbounded growth under extreme class imbalance. For Focal Loss this factor is the per-sample difficulty modulation $(1-p_t)^\gamma$, which is inherently confined to $[0,1]$ rather than a class-frequency weight; LWCE instead compresses the class-frequency weight itself.

Loss	Frequency	Difficulty	Param.-free	Bounded weight
WCE	✓	–	✓	–
Focal Loss	–	✓	–	✓
CB Loss	✓	–	–	✓
LWCE (ours)	✓	–	✓	✓

3. Proposed Method

We study a single family of re-weighting losses built on logarithmic weight compression. All three members replace the inverse-frequency weight of WCE with a logarithmically compressed counterpart and differ only in how that base weight is subsequently shaped. The three formulations and the distinct role of each are summarized in Table 3 and defined precisely below.

3.1. The logarithmic re-weighting family

Consider a classification problem with C classes, where class c contains n_c training samples. All three losses re-weight the standard cross-entropy objective by a class-dependent factor w_c,

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} w_{y_i}\ \log p_{i,y_i}, \tag{2}$$

where p_{i,y_i} is the predicted probability of the true class y_i of sample i and N is the number of samples; the members differ only in the choice of w_c. Log-Weighted Cross-Entropy (LWCE) uses the parameter-free weight

$$w_c^{\text{LWCE}} = \frac{1}{\log(1+n_c)}. \tag{3}$$

Power-Log Weighted Cross-Entropy (PLWCE) raises the logarithmic denominator to a single exponent $\alpha > 0$,

$$w_c^{\text{PLWCE}} = \frac{1}{\big(\log(1+n_c)\big)^{\alpha}}, \tag{4}$$

which continuously tunes how strongly rare classes are emphasized and reduces to LWCE at $\alpha = 1$. ε-Smoothed Log-Weighted Cross-Entropy (ES-LWCE) adds a small constant $\varepsilon > 0$ to the denominator,

$$w_c^{\text{ES-LWCE}} = \frac{1}{\log(1+n_c)+\varepsilon}, \tag{5}$$

which caps the weight at $1/\varepsilon$.

Table 3
The proposed logarithmic re-weighting family. n_c denotes the number of training samples in class c; $\alpha > 0$ and $\varepsilon > 0$ are the only free parameters, belonging to PLWCE and ES-LWCE respectively.

Loss	Class weight w_c	Role
LWCE	$\dfrac{1}{\log(1+n_c)}$	Parameter-free base weight
PLWCE	$\dfrac{1}{\big(\log(1+n_c)\big)^{\alpha}}$	Exponent α tunes emphasis strength
ES-LWCE	$\dfrac{1}{\log(1+n_c)+\varepsilon}$	Constant ε bounds and smooths the weight

3.2. Theoretical properties

We now make precise the sense in which logarithmic compression tames the weight explosion of WCE. Throughout, let $\rho = n_{\max}/n_{\min} \geq 1$ denote the imbalance ratio between the most and least frequent classes, and let the subscripts min and max index the rarest and the most frequent class, respectively. Four properties characterize the proposed family.

Proposition 1 (Monotonicity). For LWCE, PLWCE, and ES-LWCE, the weight w_c is strictly decreasing in the class count n_c. Consequently a rarer class always receives a larger weight, which preserves the qualitative behaviour of inverse-frequency weighting. *Proof.* $\log(1+n_c)$ is strictly increasing in n_c, so its reciprocal, any positive power of its reciprocal, and the reciprocal of its ε-shift are all strictly decreasing in n_c. □

Proposition 2 (Boundedness). For $n_c \geq 1$ the individual weights are bounded: $0 < w_c^{\mathrm{LWCE}} \leq 1/\log 2$ and $0 < w_c^{\mathrm{PLWCE}} \leq 1/(\log 2)^{\alpha}$. The ES-LWCE weight is bounded even without this assumption: $0 < w_c^{\mathrm{ES\text{-}LWCE}} \leq 1/\varepsilon$ for every $n_c \geq 0$. *Proof.* Each weight is maximized at its smallest admissible count by Proposition 1. For LWCE and PLWCE this is $n_c = 1$, giving $1/\log 2$ and $1/(\log 2)^{\alpha}$. For ES-LWCE the denominator is at least ε, attained as $n_c \to 0$, so the weight never exceeds $1/\varepsilon$. □

This proposition also clarifies the distinct role of ES-LWCE. LWCE is already bounded for every class that appears at least once; what ES-LWCE adds is robustness at $n_c = 0$, where the LWCE denominator $\log(1+n_c)$ vanishes and the weight diverges. Such vanishing counts arise, for instance, when class frequencies are estimated per mini-batch. ES-LWCE turns the implicit bound of LWCE into an explicit, user-controlled cap of $1/\varepsilon$.

Proposition 3 (Dynamic-range compression). The rare-to-frequent weight ratio grows *linearly* in ρ under WCE but only *logarithmically* under the proposed family. Under WCE,

$$\frac{w_{\min}^{\mathrm{WCE}}}{w_{\max}^{\mathrm{WCE}}} = \frac{n_{\max}}{n_{\min}} = \rho, \tag{6}$$

whereas under LWCE and PLWCE,

$$\frac{w_{\min}^{\mathrm{LWCE}}}{w_{\max}^{\mathrm{LWCE}}} = \frac{\log(1+n_{\max})}{\log(1+n_{\min})} = O(\log\rho), \qquad \frac{w_{\min}^{\mathrm{PLWCE}}}{w_{\max}^{\mathrm{PLWCE}}} = \left(\frac{\log(1+n_{\max})}{\log(1+n_{\min})}\right)^{\alpha} = O\big((\log\rho)^{\alpha}\big). \tag{7}$$

Comparing Eqs. (6) and (7), the compression reduces the dynamic range of the weights from linear to logarithmic in ρ while preserving the ordering that favours rare classes. Note that this is a statement about the *ratio*: unlike the individual weights of Proposition 2, the ratio is not constant-bounded but grows without limit as $\rho \to \infty$—only much more slowly than the linear growth of WCE.

Proposition 4 (Gradient-scale control). In weighted cross-entropy the per-sample gradient equals the unweighted cross-entropy gradient scaled by the class weight w_{y_i} (Eq. (2)). Hence the ratio between the minority and majority gradient magnitudes inherits the weight ratio of Proposition 3: it is $O(\rho)$ under WCE but $O(\log\rho)$ under LWCE. Logarithmic compression therefore prevents minority gradients from overwhelming the majority under extreme imbalance, which is the mechanism by which LWCE removes the optimization instability of WCE while ES-LWCE additionally caps the worst-case gradient scale at $1/\varepsilon$.

4. Experiments

5. Conclusion

CRediT authorship contribution statement

Seung Ho Go: Conceptualization, Methodology, Software, Writing – original draft. **Hyunseok Lee:** Methodology, Validation, Writing – review & editing. **Jin Hee Yoon:** Supervision, Conceptualization, Writing – review & editing.

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